STATS 403 (Spring'25) Homework 2

! For each problem, please clearly show your reasoning and write all the steps.
G As data scientists, you should feel free to google it whenever you see something unfamiliar.
☺ Group discussion for the homework is highly encouraged, but you have to write your answer by yourself. Also, you are always welcome to discuss the problems with me.

Problem 1. CNN (20')

Complete the following table. Show your reasons (instead of using PyTorch to get the answer). All the conv filters are 0 padding with stride 1.

- conv1: 32 filters with kernel size 9×9
- conv2: 64 filters with kernel size 5×5
- conv3: 64 filters with kernel size 5×5
- pool1, pool2, pool3: kernel size 2×2 with stride 2.

layer	activation dimensions	number of weights	number of bias
input	$128 \times 128 \times 3$	0	0
conv1			
pool1			
conv2			
pool2			
conv3			
pool3			
flatten	–	–	–
FC	10		

Problem 2. RNN (20')

1. Consider a function f as plotted in Figure 1. Sketch the graph of $f^{100}(x) := f \circ \cdots \circ f(x)$ where f is composed 100 times. Explain why I put such a question under the RNN section.
2. Consider the BPTT example on Slide Pages 8–10 of Lecture 6. Derive the BPTT for $\nabla_U L$.

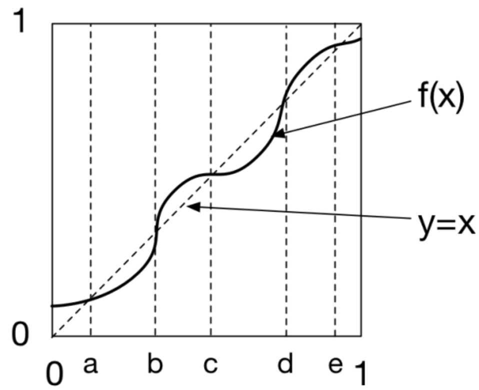

Figure 1: The graph of the function f.

Problem 3. c-transform (20')

Define

$$\phi^c(\boldsymbol{y}) = \inf_{\boldsymbol{x}} \left[c(\boldsymbol{x}, \boldsymbol{y}) - \phi(\boldsymbol{x})\right] \ ,$$

where $c(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|$. Prove that the Lipschitz norm of ϕ^c is equal to one. That is,

$$\|\phi^c\|_{\mathrm{Lip}} = 1 \ .$$

Problem 4. Lipschitz Norm and Gradient (20')

Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is a continuously differentiable function that satisfies $\|f\|_{\mathrm{Lip}} \leq 1$. Is it true that for any $\boldsymbol{x} \in \mathbb{R}^d$,

$$\|\nabla f(\boldsymbol{x})\|_2 \leq 1 \ ?$$

If true, prove it; otherwise, give a counter example.

Problem 5. DCGAN (10'+10'+5' bonus)

In class we went over the PyTorch tutorial for DCGAN.

1. Modify the codes to implement the DCGAN for the Fashion MNIST dataset. Note that you will need to change the structure of the networks. Show some samples from your generative model. There are some structures for Fashion MNIST e.g. here, but you are recommended to come up with your own.

2. Change the loss function to the one used in Wasserstein GAN and implement it using weight clipping for Fashion MNIST. Also show some samples from this model.

3. (optional) Choose any f-divergence (other than JSD) and implement an fGAN for Fashion MNIST. Specify your choice of f and show how you derived the loss. Also show some samples from this model.

Problem 6. Transformers (10'+10'+5' bonus)

Using the colab in Canvas for AttentionTXT, create a low dimensional embedding for the text provided below. Use this emedding to answer questions about Edison and Tesla. The original text is here. Stemmed version of the text is provided below. You can simply cut and paste the text as a string into the colab example.

One possible setting of parameters is SEQ_SIZE = 5, EMBEDDING_DIM = 2, HIDDEN_DIM = 10. You may try other parameters if you like.

Your task is as follows:

1. Using the word embedding matrix, calculate the distance between tesla and edison to other words in the text, and use these distances to answer the following questions:
   A. Who of the two persons (Edison or Tesla) was more scientific?
   B. Who was a better inventor?
   C. Who was a better developer?
   D. Who researched more about electricity?
   E. Who worked on incandescent light?
   F. Who worked more on motor design for alternating current?
   The stems of relevant words are incandesc, electr, inventor, gener, develop, light, scientif, motor, altern
   Hint: the embeddings give coordinates of the words in the embedding space, so you can find the coordinates for Tesla and Edison and use them as reference points to calculate distance to other words.

2. Repeat the task of question 1, but this time remove the transformer part in the model and replace it with linear NN with the same embedding and hidden parameter values. Using the new embedding, answer again the questions A-F. Are there any meaningful differences between the answers?

3. (optional) You may also visualize the embedding by plotting it, and use the graph to explain your answers. The visualization code is provided below.

```
import matplotlib.pyplot as plt
Embed = torch.Tensor(model.embed.weight).detach().numpy()
type(Embed)
fig, ax = plt.subplots()
ax.scatter(Embed[:,0], Embed[:,1])
for i, label in enumerate(vocab[:]):
    ax.annotate(label, (Embed[i, 0], Embed[i, 1]))
```

The stemming version of the text is provided here:

thoma edison is wide known as one of histori most consequenti inventor a legaci born of both hi indisput geniu in the laboratori and note ruthless as a businessman; the old adag is that histori is written by the victor and in edison case he spun hi narr in real time sometim as in the case of invent develop by contemporari such as nikola tesla that meant bend the truth a littl bit born in 1847 at the veri end of the industri revolut edison wa part of a new wave of scientist and inventor that lit the way into the modern era hi fame research lab in menlo park new jersey host the develop of innov that still undergird much of our industri and consum infrastructur includ the phonograph (which record and play sound) motion pictur and the light bulb hi work there wa so import that the town in which menlo park wa locat now bear hi name obsess with hi work and known to be an exact boss edison had an ego as incandesc as hi light bulb a sens of hi own great that wa undoubtedli justifi he wa also incred competit will to do whatev wa requir to ensur that hi idea won out born in serbia tesla wa a differ kind of geniu wherea edison wa an etern experiment and tinker tesla wa a human calcul and hi abil to work out complex math and physic equat in hi mind help him achiev earli career success in europ after a nomad adolesc spent travel and take class across eastern europ tesla wound up in hungari at the age of 25 hire to work as an electr engin at the budapest telephon exchang he excel there channel the workahol tendenc he'd display as a top student back in croatia befor drop out of school within a year he wa off to pari to work for the continent edison compani an offshoot of the inventor success american busi at that point electr wa begin to light up the street of citi around the world at first most citi were use high voltag arc lamp to illumin the night sky but while they shone bright and amaz a civil that had throughout all of histori been govern by the sun rise and fall the earli light technolog also present a problem: it wa veri danger arc light wa fuel by power station that pump through more than 3000 volt of electr at a time which often led to spark overh and full on explos in public place rain flicker of electr down on pedestrian and start fire with regular electr gener by direct current (dc) wa a far safer altern and onc edison develop a stabl and long last incandesc lightbulb he set out to provid light to home and build around the world by 1882 hi edison illumin compani open the world first central power station on pearl street in manhattan the compani use direct current to deliv 110 volt of electr to nearbi build it start out with 59 custom and provid a significantli reduc risk of accident mishap for the next two year edison dc electr gener spread the incandesc light to a grow number of citi across the countri but for it grow reach it had a signific weakness: electr deliv by direct current could onli travel so far especi in those earli day as a result other inventor continu to develop what wa call altern current ac which could easili modul voltag use transform tesla wa one of those ac devote in 1882 while work for edison parisian outpost and out on a walk with a friend tesla wa suddenli struck with the solut to an engin challeng that had been vex him for some time hi abil to visual entir complex mathemat equat and feat of engin came in handi as tesla dream up the mechan of an ac gener motor develop and hone the altern current gener becam someth of an obsess for him and in 1884 when hi american boss in pari wa summon back to the unit state he suggest that tesla emigr to the west and so in septemb of that year tesla arriv in manhattan after a harrow journey in which most of hi possess were stolen dure a mutini on the ship howev money wouldn't be a concern for all that long tesla ran into edison himself who invit the immigr inventor to work for him at the edison machin work on the lower east side of manhattan a letter of recommend from hi old boss did not hurt hi caus at the time tesla recal in an interview year later he wa in awe of hi new boss "thi wonder man who had receiv no scientif train yet had accomplish so much fill me with amazement" tesla wa put to work on a varieti of project includ repair the circuitri system on the oregon the first boat to be lit by electr power reassembl dc gener and other task year later in an interview conduct in 1921 tesla recal impress edison with hi quick fix of the oregon light system to the point that edison declar him a "damn good man" the serbian inventor wa also charg with creat an arc light system but hi usag of ac power wa of no interest to edison who had a fortun invest in dc power and did not want to lose hi loyalti later tesla would purportedli say that edison himself promis a larg sum to improv on the dc system then retract the offer when tesla present him with hi work claim that he "did not understand american humor" infuri tesla to the point that he storm out and set out on hi own determin to spite the elder inventor tesla onli work for edison for about six month and after a time spent dig grave he receiv enough investor cash to set up hi own compani in rahway new jersey close to menlo park those investor took the compani out from under him and it wasn't until 1887 with a new factori in manhattan that tesla wa abl to truli pursu hi ac motor it wouldn't take long befor he'd master the machin as he wa award seven separ patent for it variou mechan in the spring of 1888 soon after he licens those patent to georg westinghous edison chief rival in the race to suppli citi with power the race between ac and dc would escal from there with edison pull out nearli all the stop to prove that ac wa danger (though the fame death by electrocut of topsi the eleph wa not hi doing) to discredit what would ultim prove a far superior system tesla die nearli penniless though that had noth to do with edison he made a fortun from hi contract with westinghous but lost it all through poor busi deal bad invest and expenditur on grand experi that result in failur hi suppos rivalri with edison wa almost by associ and differ of opinion on scientif matter highlight onli in hindsight as tesla incred career which span far more than ac power innov

Problem 7. Shapley Values (20' bonus)

A cooperative game specifies for every group of players what is the total payoff that the members of a group (coalition) S can obtain by signing an agreement among themselves. The payoff for coalition S is $v(S)$. We also assume that by forming a coalition, the payoff for the coalition as a whole is higher than the sum of payoffs to individual members or sub-groups in that group. The coalition payoff S can be considered as the total amount that can be distributed between its members.

An agreement between the players $x_1, x_2, \cdots, x_N$ is the amount (marginal contribution) that each individual member should receive based on a claim (demand for payoff) that the player has both individually and as part of all possible coalitions.

To find the marginal contribution, *Shapley* proposed the following scenario: the players enter into a room to demand the payment in different order. Depending on the order, each one will be paid slightly differently. The fair solution is the one that is the average over all possible orders of arrivals. Lets give an example:

Suppose that there are two players and $v(\{1\}) = 10$, $v(\{2\}) = 12$ and $v(\{1, 2\}) = 23$. There are two possible orders of arrival: (1) first 1 then 2, and (2) first 2 then 1. If 1 comes first and then 2, 1's contribution (payoff) is $v(\{1\}) = 10$; when 2 arrives the surplus value increases from 10 to $v(\{1, 2\}) = 23$ and therefore 2's marginal contribution is $v(\{1, 2\}) - v(\{1\}) = 23 - 10 = 13$. If 2 comes first and then 1, 2's contribution is $v(\{2\}) = 12$; when 1 arrives the surplus increases from 12 to $v(\{1, 2\}) = 23$ and therefore 1's marginal contribution is $v(\{1, 2\}) - v(\{2\}) = 23 - 12 = 11$. The fair payoffs to individuals are then calculated by averaging between the two cases. Thus 1's expected marginal contribution is: $(10 + 11)/2 = 10.5$, and 2's expected marginal contribution is $(13 + 12)/2 = 12.5$. This is the *Shapley value*: $x_1 = 10.5$ and $x_2 = 12.5$.

<u>Exercise</u>:
Suppose that there are three players now and $v(\{1\}) = 100$, $v(\{2\}) = 125$, $v(\{3\}) = 50$, $v(\{1, 2\}) = 270$, $v(\{1, 3\}) = 375$, $v(\{2, 3\}) = 350$ and $v(\{1, 2, 3\}) = 500$. Write a table where each row has 4 columns: order of arrival, 1's contribution, 2's contribution, 3's contribution. For example, the first row can be something like:

first 1 then 2 then 3 (123) | $v(\{1\}) = 100$ | $v(\{1, 2\}) - v(\{1\}) = 170$ | $v(\{1, 2, 3\}) - v(\{1, 2\}) = 230$.

Complete the remaining rows and write down the complete table. Consider all rows as equally probable and use the table to calculate the Shapley value for each player x_1, x_2, x_3. Does the sum of marginal contribution exceed, is less or equal to the value of the complete coalition ($v(1, 2, 3)$)?

Problem 8. Normalizing Flows (20')

Train a Real-NVP using the MNIST dataset and then use it as a generator. You need to show your codes and some generated examples.

Problem 9. Diffusion Models (20')

It is known that if $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$ and $p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}|\mathbf{A}\mathbf{x} + \mathbf{b}, \mathbf{L}^{-1})$, then

$$p(\mathbf{x}|\mathbf{y}) = \mathcal{N}\left(\mathbf{x}|\boldsymbol{\Sigma}\left(\mathbf{A}^{\mathrm{T}}\mathbf{L}(\mathbf{y} - \mathbf{b}) + \boldsymbol{\Lambda}\boldsymbol{\mu}\right), \boldsymbol{\Sigma}\right),$$

where $\boldsymbol{\Sigma} = (\boldsymbol{\Lambda} + \mathbf{A}^{\mathrm{T}}\mathbf{L}\mathbf{A})^{-1}$. Use this fact to show that in a diffusion model,

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

can be written as the Gaussian

$$\mathcal{N}\left(\mathbf{x}_{t-1}\middle|\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t}\mathbf{x}_0, \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}\boldsymbol{I}\right).$$

Problem 10. NN-GP (20')

Under the settings of our lecutre on NN-GP, show that

$$K^l(x, x') = \sigma_b^2 + \sigma_w^2 \mathbb{E}_{z_i^{l-1} \sim \mathcal{GP}(0, K^{l-1})}\left[\phi(z_i^{l-1}(x))\phi(z_i^{l-1}(x'))\right].$$

Explain why the inductive relation is true that

$$K^l(x,x') = \sigma_b^2 + \sigma_w^2 F_\phi\left(K^{l-1}(x,x'), K^{l-1}(x,x), K^{l-1}(x',x')\right)\ .$$